

Model-driven multicomponent volume exploration

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Abstract The current multicomponent volume segmentation and labeling methods are mostly hard to get correct segmentation and labeling results automatically and rely hardly on experts' aids, which make related volume exploration to be time consuming, laborious and prone to errors and omissions. To solve this problem, we present a novel volume exploration method driven by admitted model. We first apply Gaussian mixture models to segment the raw volume. However, different components with similar value are still mixed. To segment these components further, we make use of region-grown principle to produce a fine-grained part segmentation. To label different parts automatically, we found that it is helpful to take advantage of annotated model, like human anatomy model (PlasticboyCC, <http://www.plasticboy.co.uk/store/index.html>, 2013). However, it is not straightforward to label segmented volume with geometric model automatically. Inspired by electors voting (Au et al., Comput Graph Forum 29:645–654, 2010), we propose a volume-model correspondence schema to overcome this intractable challenge. Moreover, it is essential to exploit intuitive interactive methods for interactive exploration, so we also developed practical precise interaction techniques

to assist volume exploration. Our experiments with various data and discussion with specialists show that our method provides an efficient and impactful way to explore volume data.

Keywords Interactive volume visualization · Volume segmentation · Volume correspondence · Knowledge-assisted visualization

1 Introduction

Three-dimensional volume data are widely used in many fields such as medicine. These data are often complex and composed of many components. Segmentation and labeling the components in a way that they can be readily distinguished are often critical for helping users understand and analyze the data. However, the visual selection and visualization of relevant semantic information [5,9,17,34,35] is a tedious and daunting task, as it often requires users to try many different combinations of segmentation, annotation and color maps based on transfer functions, which makes related volume exploration to be time consuming, laborious and prone to errors and omissions. As a result, how to segment and label multicomponent volume data automatically and provide necessary interaction means is still a challenge of great interest.

Some pioneering studies have considered specifying semantics for multicomponent volume rendering from particular viewpoints. For example, before labeling the components, the user is always required to design a transfer function or perform a segmentation to classify regions [7,8,27,35,37,40]. After segmentation, transfer functions [10,11,13,14,18,24,33] can be defined to provide a rendering which displays a lot of useful information, and

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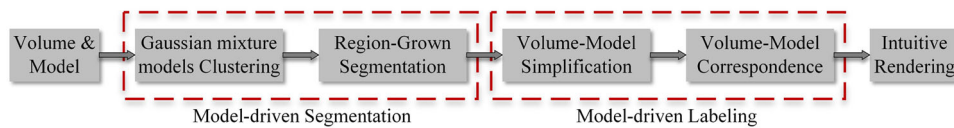


Fig. 1 The major steps of our approach to explore multicomponent volume driven by model. The semantic reference model incorporates expert knowledge about specific annotation and appropriate rendering, and is utilized to help segmentation, labeling, and rendering processes

annotations may further improve the output, as noted in [2, 5, 23, 25]. However, there are a number of potential pitfalls with these methodologies. First, before suitable annotation can be added, the user must find out scalar values for the hidden structures to be classified. Yet, without examining the data first, the user may not have a clear idea of what structures are presented and what they are, even in general terms: such judgment requires expert domain knowledge. Any initial guess is unlikely to be able to classify hidden features successfully, and so the labeling algorithm in turn will be unable to provide proper annotations. Also, as these methods require input in the form of a transfer function or segmentation, if the user decides the current output is unsuitable, and wishes to try another possibility, either resegmentation must be performed, or other parts of the pipeline must be recomputed to determine the labeling. Doing so iteratively can take a long time and makes exploring transfer functions interactively impossible. Second, many structures in 3D volume data must consider multiple scalar values for appropriate segmentation and rendering. They may require prior information (total segment numbers or even color attribute metrics), with multidimensional transfer functions. Given these inherent shortcomings, purely scalar-based segmentation, labeling or rendering algorithms can be limited for practical use.

As a solution to this problem, efforts have begun to store visualization expertise, perhaps incorporating semantic models, to help provide users guidance during data visualization, suggesting to the user both labeling information and the best rendering parameters for the data and task at hand. This is an immensely useful and promising development, saving the user from having to rediscover or reinvent good solutions [35].

In this paper, we propose a novel volume exploration method that segments and labels multicomponent volume data driven by a semantic model, like human anatomy model (HAM). The user could create a sensible and readily usable labeled and rendered volume by analogical inference from the given semantic model. Both novitiates and experts could explore their volume data intuitively with our precise volume interaction techniques. The overview of our approach is shown in Fig. 1. First, we apply Gaussian mixture models (GMM) to segment the raw volume. However, different components with similar value are still mixed. To segment these components further, we make use of region-grown principle to produce a fine-grained part segmentation. To label differ-

ent parts automatically, we found that it is helpful to take advantage of annotated model, like HAM [32] as shown in Fig. 2. However, it is not straightforward to label segmented volume with geometric model automatically. Inspired by electors voting [3], we proposed a volume-model correspondence schema to overcome this intractable challenge. Moreover, it is essential to exploit intuitive interactive methods for user's interactive exploration, so we also developed practical precise interaction techniques to assist volume exploration.

Our main contributions are as follows:

- We propose a novel multicomponent volume exploration framework driven by model.
- We construct an explicit volume segmentation by applying GMM integrated with region-grown principle.
- We present a volume-model correspondence schema by volume and model simplification and adapted electors voting method.
- We assist the exploration with essential intuitive interactive visualization techniques with precise real-time feedback information.
- We demonstrate the applicability of our approach in several applications and discussions with experts.

The rest of this paper is organized as follows: The related work is summarized in Sect. 2. An overview of our method is presented in Sect. 3. Section 4 discusses technical details, including segmentation, labeling, and interactive rendering, guided by the semantic model. Results, performance analysis, and discussions with experts are demonstrated in Sect. 5. Finally, conclusions and future work are presented in Sect. 6.

2 Related work

Related work falls into three categories: (1) interactive volume data segmentation and annotation, (2) correspondences between multiple components, and (3) knowledge-assisted rendering, which we overview next.

2.1 Segmentation-related volume rendering

As noted in [37], segmentation is an essential component of high-quality volume-based rendering. Tzeng et al. [40]

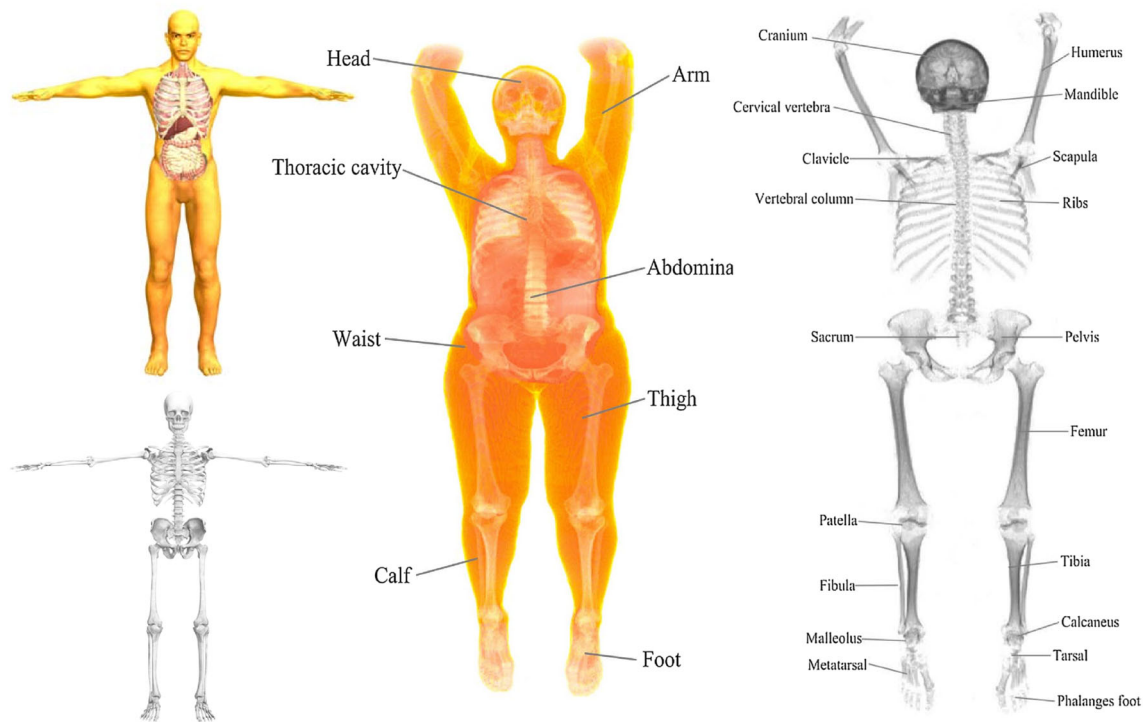


Fig. 2 Snapshot of our method. *Left* semantic full body model with internal skeleton used to drive visualization. *Center, right* a full human exploration derived from raw volume

developed an interactive user interface for specifying classification functions by allowing users to paint directly on sample slices of the volume. So, Chen et al. [8] introduced a seeded region-grown volume segmentation system, which extended strokes and sweeps through the volume, relieving users of the need to browse through hundreds of cross-sectional slices. Inspired by the volume catcher method [28], Yuan et al. [45] proposed a method for extracting 3D volume structures based on simple strokes drawn directly on rendered images. With VolumeShop [5], Bruckner and Groller proposed an interactive system for direct volume illustration. Based on the initial classification, our volume segmentation further automatically segments the classified regions into parts using a region-grown scheme, and provides intuitive tools to refine the segmentation results.

Related atlas-based segmentation usually requires the use of image registration to align the atlas image or images to a new, unseen image. In [26], an atlas of the rabbit brain based on magnetic resonance imaging is presented for the automatic parcellation of the rabbit brain. Since atlas-based segmentation is highly dependent on the registration framework used to guide the atlas label propagation, [44] provides a single-step and three two-step registration methods based on affine and deformable registration algorithms in the ANTS and FSL algorithms. Cabezas et al. [6] provides a comprehensive review of atlas-based segmentation.

Most of the present segmentation-related methods make use of data value like density or derived values like gradient. These methods could segment tissues with different data values. As these works do not automatically analyze data with position information or admitted models, most of present methods could not segment same tissue at different position.

2.2 Annotation and correspondence

Annotation and labels provide legends to establish relations between pictorial elements and textual expressions [1]. Adding notation to multicomponent 3D models [2, 4, 21–23, 30] requires methods similar to image annotation techniques [29], which can be classified into manual annotation, semi-automatic annotation, and automatic annotation approaches.

In volume visualization, manual annotation has been applied to recreate the appearance of static illustrations [5, 25] and simplify recognition difficulties [19]. Manual annotation is considered “best case” in terms of accuracy, since labels are selected based on human determination of the semantic content. Nevertheless, at the same time, it is a tedious and monotonous process.

A strategy of semi-automatic annotation is better than manual annotation in terms of efficiency and better than automatic annotation in terms of accuracy as reported by [43] in an image annotation survey. Automatic labeling bridges the

gap between the low-level components and high-level human semantics, and can be based on machine learning [23], prior knowledge of specific artificial objects [21], or characterization of components [2]. Unlike these approaches, our system takes cues by comparing existing semantic models to the input data: correspondences [22, 39] between the model and the segmented volume data are derived by spatial correlation between components.

Three-dimensional multicomponent volume data comparison is a challenging problem because models and input volume contain significant structure at many different resolutions. All existing 2D/3D scene comparison algorithms find attempting to compare the finest-level data directly is rarely successful [15]. Instead, the data are often transformed into a representation that enables the comparison of important features. Several attempts have been made at comparing 3D scenes. One approach works by forming a graph by connecting adjacent regions [31]. A feature vector for each graph is constructed using the eigenvectors of the spectral decomposition of the graph topology. The construction of this graph relies on Reeb graph theory and has only been tested on scenes with up to less than ten components. For multicomponent artificial scenes, Fisher et al. [15] represented scenes as graphs that encode models and their semantic relationships, then defined a kernel between these relationship graphs that compares common virtual substructures in two graphs and captures the similarity between their corresponding scenes. Their algorithm is not directly applicable to our current situation in which the input data are unlabeled. By making use of the skeleton, [3] proposed an automatic unlabeled shape correspondence algorithm using a voting scheme whereby electors are quickly selected through a combinatorial search. Enlightened by [3], we also transform scenes into a relationship graph, and use a revised electors voting method to find the best combinatorial match between two sets of graph nodes.

2.3 Knowledge-assisted rendering

As a central topic in direct volume rendering, choice of transfer function (TF) has been widely studied [13, 14]. By mapping the volume values to colors and transparencies, the TF distinguishes the desired features from background in the rendered image. Zhou and Takatsuka [46] utilized topological attributes derived from the contour tree of a volume to perform automatic TF generation. The contour tree provides a visual index to volume segments, and allows discovery of global features in volume data. Ruiz et al. [36] proposed an automatic approach for TF design without requiring prior knowledge or pre-segmentation information, but it needs the user to provide a target distribution.

The technique presented here is related to knowledge-assisted visualization techniques [9, 19, 27, 35], which aug-

ment rendering with expert knowledge to make it easier for the casual user and even for domain experts. Major challenges are how to collect this knowledge, how to select it, and how it is used by the given task. For instance, Rauteck et al. [34, 35] presented two similar semantics-driven visualization frameworks that enabled users to specify a mapping from multiple data features to visual attributes via textually formulated rules, allowing them to define explicit semantics for attributes of interest. A further semantic method proposed by [17] combines the advantages of a graphical specification of mappings with the illustration capabilities provided by a rule-based approach. However, these papers work on the transfer function space rather than applying the semantic information directly to the geometric volume data. Unlike these, we make use of color information directly from the model to get a reliable illustrative rendering.

Correa et al. [10] showed how to define size-based transfer functions that could improve classification and enhance volume rendering results. The same authors [12] proposed a semiautomatic method for generating transfer functions that progressively explore the transfer function space to maximize visibility of important structures. In [33], Pražni took the shape of volume features into account to classify different regions of volume data. In [42], Wang et al. introduced a novel volume exploration scheme, which approximates the exploration space using an incremental Gaussian function to ease classification of time-varying data. We also use GMM to aid segmentation, but the user's input is combined with the corresponding semantic model. The original classification separates the input data into different regions with varying properties, and then this coarse segmentation must be further refined into parts. Taking the foot as one example, an initial classification separates the bones as a whole from the data, and then the individual bones must be further segmented as tarsal or metatarsal.

In [20], a new ROI-based MM visibility-driven transfer function for PET-CT visualization is proposed. The method uses so-called “visibility” metric to present how much of the ROIs are visible to the users. The metric also measures the visibility of the ROIs in PET in relation to how it is affected by transfer function manipulations to its counterpart CT. An intuitive ROI selection tool based on automated PET segmentation is provided for ROI selection. Friese et al. [16] presented a new software to visualize and segment three-dimensional medical datasets based on fast and interactive volume visualization.

3 Overview

Figure 1 sketches the major steps of our approach. Our system is based on a key idea: that a reasonable guide to segmentation, labeling and color mapping can be inferred

from the information in an annotated reference model. Doing so reduces the degree to which the user needs to be a domain expert, or visualization expert. The provided semantic model is exploited to help volume data segmentation, labeling, and rendering. In our experiments, our reference models were based on models (including male and female anatomy packs) from the Plasticboy anatomy models store [32].

A typical 3D medical volume dataset is a group of 2D slice images acquired by a computed tomography (CT), magnetic resonance imaging (MRI), or MicroCT scanner. Each voxel of the data is represented by a single value obtained by sampling the immediate area surrounding the voxel. In our method, the volume data are transformed into the resulting visualization with annotated information after three main steps: segmentation, labeling and rendering, as follows:

- *Model-inspired segmentation*: The input volume data are semi-automatically segmented using a classification–decomposition–interaction scheme, guided by the model. Initially using a number of segments determined by the semantic model, automatic coarse classification is done using GMM, and a region-grown method is used to improve the segmentation. The user can further refine the segmentation results using simple sketch-based splitting and uniting operations by comparison with the given model.
- *Model-driven labeling*: Components of the segmented volume are automatically annotated by inference from the corresponding labeling information in the reference model. This is done automatically using correspondences built by spatial component correlation. The user can revise these correspondences by simple selection of components.
- *Intuitive rendering*: The labeled components are automatically rendered with appropriate colors adopted from corresponding coloring in the reference model. The user could adjust the color mapping further. Other intu-

itive precise interactive visualization techniques are constructed to assist user's further analysis.

Overall, the user could explore their multicomponent data with an accurately segmented and labeled model, which is realistically rendered, by analogical inference from the provided model.

4 Method

Next, we describe our method in detail including segmentation, labeling, and rendering.

4.1 Segmentation

The challenge of visualization of a complex 3D volume is magnified by the difficulty of accurately segmenting volume data. We propose a semi-automatic segmentation approach, driven by the input semantic reference model. Our segmentation approach is based on a classification–decomposition–interaction scheme (see Fig. 3). First, voxels with similar scalar values are grouped using GMM. Using each of these classes as input, a region-grown method is applied to further segment the volume data. Finally, interactive sketching may be used to split and unite segmented components with real-time feedback as needed, and local details may be refined. It is necessary to attend the proposed segmentation method which could not only segment different tissues, but also divide different parts of the same tissue that is critical.

First, we explain how classification is performed by GMM working in 2D feature fields (scalar and gradient magnitude). This performs a maximum likelihood feature separation of the feature space to estimate the underlying probability density functions (pdfs). Unlike [42], the number k of mixture components is suggested by the reference model, which stores structure information. In practice, number of k in GMM is automatically assigned with the number of different tissues in corresponding parts of the model. The

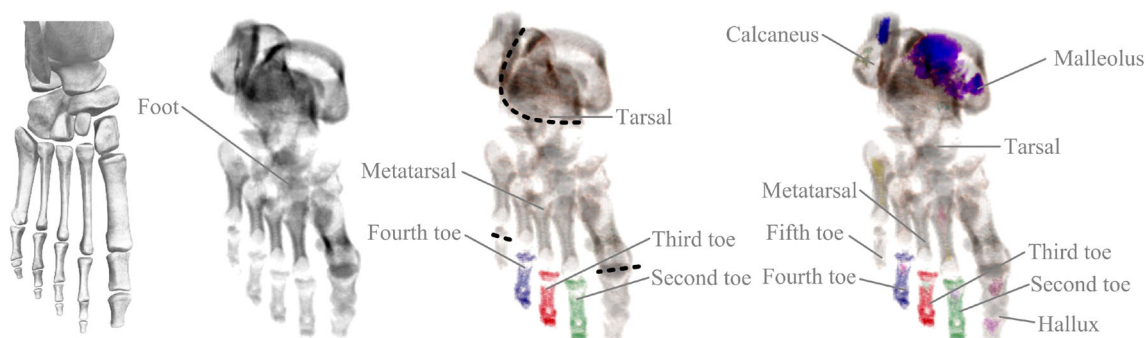


Fig. 3 Semi-automatic segmentation: *left* provided model; *mid-left* classification separating the bones from the foot volume; *mid-right* automatic decomposition with splitting operation cues (*dashed curves*); *right* final segmentation after interaction

conditional pdf of the observation vector with respect to the different classes is modeled as a linear combination of multivariate Gaussian pdfs. Each homogenous region in the 2D feature space is represented by a Gaussian distribution

$$g(x|\mu, \Sigma) = \frac{1}{2\pi|\Sigma|^{1/2}} e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)} \quad (1)$$

where x is a vector in the feature space, μ is the mean vector of the Gaussian, and Σ is the 2×2 covariance matrix. As such, the distribution of all k regions is represented by the Gaussian mixture distribution

$$p(x|\theta) = \sum_{j=1}^k \alpha^j g^j(x|\mu^j, \Sigma^j) \quad (2)$$

where α^j is the prior probability of the j th Gaussian satisfying $\alpha^j = 1/k$, and θ denotes the parameter set of the GMM with k units $\{\alpha^j, \mu^j, \Sigma^j\}_{j=1}^k$. Assuming that the vectors in the 2D feature space $\{x^1, \dots, x^n\}$ are independently identically distributed, the maximum likelihood estimation of θ is

$$\hat{\theta} = \arg \max_{\theta} p(x^1, \dots, x^n) = \arg \max_{\theta} \prod_{i=1}^n p(x^i|\theta) \quad (3)$$

where n is the number of voxels in the volume. To find the maximum-likelihood estimate of the parameters of the underlying distribution, we use the well-known expectation-maximization (EM) algorithm [41] to determine θ iteratively. The E-step finds the expected value of the log-likelihood $\sum_{i=1}^n \log p(x^i|\theta)$ and the M-step finds new parameters that maximize the expectation computed in the E-step. The obtained mixture components are then converted to ellipsoidal Gaussian transfer functions (ETF) as in [42]. After this classification, voxels with similar scalar values are grouped, but not fully segmented in the volume space.

Next, a region-grown decomposition method is applied to decompose each class provided by the GMM into components. For a given class, the region-grown method automatically and progressively adds seeds as follows. A random seed is selected from the voxels, and then region-grown is performed from the seed until the distance between the boundary voxels and their neighbors is larger than a threshold (1 voxel edge length). As regions are formed, their voxels are removed from the class, and the process is repeated until voxels have been assigned to regions. Each class is thus segmented into several isolated components; we delete small components with fewer than 4 voxels.

Sketch-based interaction allows the user to correct the segmentation using two main operations (splitting and uniting) to revise these automatically detected 3D volume components where the number of parts of volume is not equal to that of corresponding parts of the model. The user should

also consider the reference model when doing so. We provided a visual comparison between a model and a volume. The partial matching result, which has accuracy less than 100 % as discussed in Sect. 5.2, will be highlighted for further confirmation. If user finds something wrong, he could easily segment corresponding region as guided with our system.

In the example in Fig. 3, any bones wrongly segmented by the region-grown method could be united by the user, based on the correct information in the provided model. User only needs to draw single line on where it needs to be split further. Our method computes curved surface through the line and perpendicular to the screen. Then, the volume is subdivided within the curved surface. Similarly, the splitting operation can be used to separate wrongly combined bones. In addition, a refinement tool is also provided as in [45], allowing the user to draw freehand curves optionally to remove unwanted voxels from the segmentation results selectively.

4.2 Labeling

Unlike lots of exiting labeling algorithms, which tab their volume or geometry by hand, we aim to label the segmented volume data automatically to release users from tedious and error-prone complex multicomponent volume labeling. We make use of semantic reference model including annotations in the form of labeling information to realize efficient and widely approbatory labeling. Therefore, it is necessary to find correct correspondence between the volume and the model. However, it is not straightforward to get these correspondences, since the concrete volume is distinctly different from the standard model in position and especially in shape. What is more, the correspondence between the volume (V) and the model (M) is mostly partial matching problem, where the volume always corresponds with part of the model. To get over of these obstacles and inspired by the electors voting method, we construct a volume-model correspondence scheme as illustrated in Fig. 4. Segmented volume comes from the prior segmentation step as discussed in Sect. 4.1. Our semi-automatic approach has five steps:

- *Graph construction*: We start by constructing related relationship graphs for M and V , denoted by $\hat{M}$ and $\hat{V}$, respectively. Volume graph with the point represents the part after segmentation, the edge between two points represents the neighborhood relationship between these two parts constructed from the segmented volume. After segmentation, the clustered volume data are abstracted to get volume graph. Model graph is similarly got from basic model. Each node has node degree and other attributes, such as value for volume and color and labeling for model.

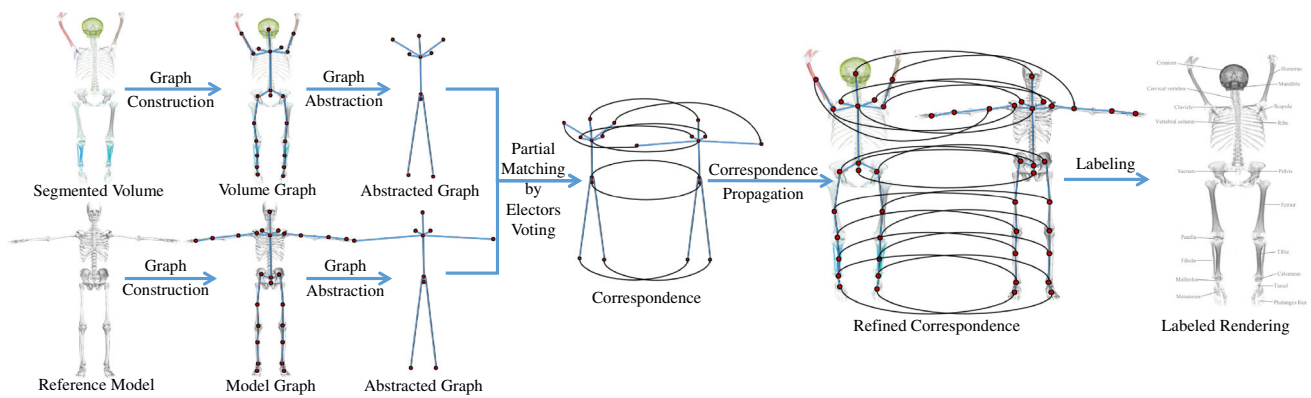


Fig. 4 The illustration of our volume labeling pipeline. Segmented volume are labeled by simplification and correspondence with basic model from left to right

- *Graph abstraction*: There are always hundreds of organs for the whole human body. The partial correspondence of skeleton with hundreds of nodes is heavily time consuming. On the other hand, we found that it is reasonable to simplify the graph by deleting nodes with degree 2, since the reverse process is direct and efficient. Similarly, the source standard complete human anatomy (CHA) model is abstracted and simplified too. We abstract the graph to a simplified version as $\hat{M}$ and $\hat{V}$. We also store the masking information (an indexed vector with 1 for a retained node and 0 for a masked node).
- *Partial matching by adapted electors voting*: By now, we have transformed the raw labeling problem to partial corresponding problem, which has been researched in years. After getting the simplified graph, we adapt an existing approach to the partial correspondence, which uses statistical method to synthesize a high quality correspondence rather than a direct search for the best correspondence using score function, namely electors voting [3]. Its key idea is to form a set of reliable electors quickly, which then note on individual feature-to-feature matching. It is assumed that the volume data of (part of) human body do not have topological change compared with corresponding model. This assumption is important for us to make valid use of electors voting.
- *Correspondence propagation*: After computation of correspondence, it is needed to refine the correspondence by reversing the simplification process. Utilizing the masking information, we undo the abstraction and propagate correspondences from the retained nodes to the masked nodes. Therefore, we establish a correspondence between volume data and CHA model.
- *Labeling*: After component correspondences between M and V have been settled, labeling information is automatically transferred from M to V .

Differing from exiting elector voting method, which deals with geometrically model-to-model correspondence prob-

lem, our adapted electors voting method replies with the volume-to-model correspondence problem. In our adapted electors voting, elector selection involves a branch-and-bound search on a combinatorial tree (see Fig. 5). Every node represents a possible component-to-component correspondence, except for the root that is an auxiliary node and represents an empty set. A path from the root to any tree node (for example, the highlighted path in Fig. 5) represents a possible correspondence solution comprising all component-to-component correspondences along the path. The tree is expanded while searching from the root. A node is added only if the new correspondence set including the new node passes a cascade of pruning tests, otherwise the subtree rooted at that node is pruned. The pruning tested are designed by easily measurable deviations of intrinsic topological information associated with pairs of components. Let (p_k, q_k) , p_k in $\hat{M}$ and q_k in $\hat{V}$, denote the component pair being considered for addition to the k th level of the search tree. The set $\Omega = \{(p_1, q_1), \dots, (p_k, q_k)\}$ denotes the correspondence set containing all component pairs on the path from the root to (p_k, q_k) .

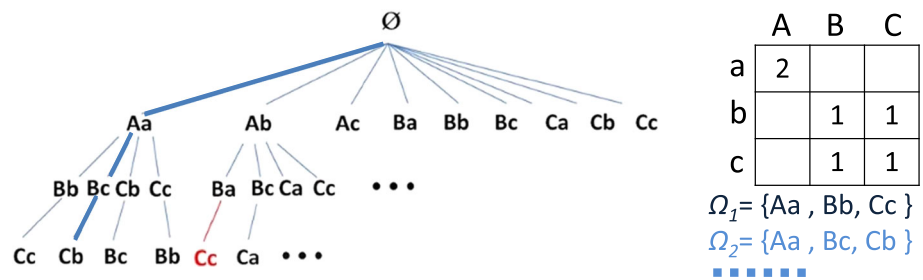
The topological test depends on the node degree difference. We reject (p_k, q_k) if:

$$\| \text{Degree}_{p_k} - \text{Degree}_{q_k} \| / \max\{\text{Degree}_{p_k}, \text{Degree}_{q_k}\} < \varepsilon_d \quad (4)$$

where $\varepsilon_d = 0.5$ for all tested volume data.

Above test efficiently filters away different types of incorrect correspondences by considering different topological information. After the search, every tree node represents an elector, which contains all component-to-component correspondences along the path from the root to that node. Every elector casts one vote for each of its constituent component pairs; the votes fill in the voting table. The component pair with the highest votes is then iteratively picked as the final correspondence. Each matching component pair (p_i, q_j) in

Fig. 5 Combinatorial search tree instance and voting table



an elector Ω_k contributes one vote. So, each element $s_{i,j}$ in the voting table S is the total number of occurrences of the component pair (p_i, q_j) in all electors:

$$s_{i,j} = \sum_k \Omega_{i,j,k}, \quad \Omega_{i,j,k} = \begin{cases} 1 & \text{if } (p_i, q_j) \in \Omega_k \\ 0 & \text{if } (p_i, q_j) \notin \Omega_k \end{cases} \quad (5)$$

However, sometimes failure cases happen with the automatic labeling process above, like labeling human head, because structure abstraction of the head is too confusing to compute a complete correct correspondence. In such condition, user needs to mend the result by specifying corresponding organ in the model of the volume parts interactively. We thus provide a correspondence tool that allows the user to fix incorrect correspondences between M and V . Such user interaction (building a correspondence by selecting p_i and q_j directly) could easily be adopted in the partial matching procedure to assist in the elector selection and voting process. During elector selection, the built correspondence would prevent other component pairs including p_i or q_j , and so would reduce the size of the search space. For example, one assigned correspondence could reduce the combinatorial tree size from $n!$ to $(n-1)!$ for $1-1$ matching for M and V , whose component numbers are both n . Assigned correspondences would also decrease the value of votes to be considered. Our experiments demonstrate that use of interaction can speed up the partial matching, and improve accuracy even with a very few interactions.

4.3 Rendering

Volume rendering uses mapping functions to convert original data into renderable scalar color and opacity values to help the user understand features of the data. The scalar value is mapped to voxel data with TF, and then displayed by evaluating an optical model, which describes how the volume emits, reflects, scatters, absorbs and occludes light. To create an appropriate TF, users need both general experience of doing so, and insight into the particular visualization problem at hand.

Unlike traditional typical transfer functions defined in a purely interactive [18,24] or automatic [36] way, we determine a TF mostly automatically from the provided

semantic model, while allowing adjustments to be made by the user. Our approach to TF definition is based on the correspondences built earlier. For all component correspondences between M and V , we directly sample the RGB color of each component in M , and assign this RGB value to the corresponding components in V . After automatic color assignment, the user could choose to modify the TF color mapping by using simple interaction with direct feedback. In addition, it is essential to assign appropriate opacity to the corresponding components. For photorealistic rendering, we found that assigning opacity to 0.7 is sufficient in most cases, which means that tissues are rendered near opaquely.

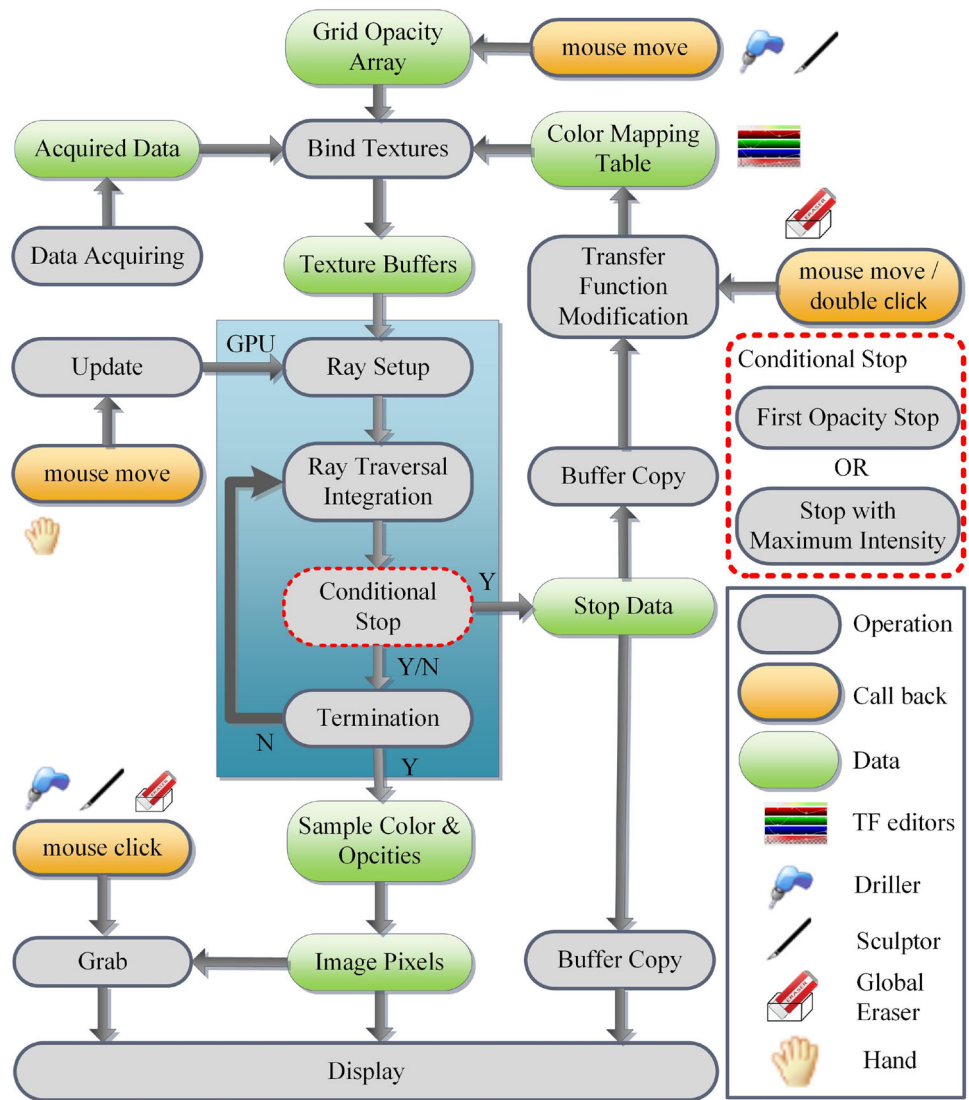
To support interactive exploration, it is necessary to construct some essential intuitive interactive visualization techniques. Recent researches indicate that sketch-based interaction techniques are much more intuitive than traditional Window, Icon, Menu, and Pointer (WIMP) based interaction. However, sketch-based techniques are generally up against precise problem. That is to say, most sketch-based techniques cannot achieve precise interaction, which is vital for visualization. Here, we present a framework of sketch-based visualization techniques with precise feedback information as shown in Fig. 6, which is a GPU-based ray casting process comprising four steps.

1. The source data and color mapping table are acquired by the CPU and bound by the GPU as textures [14].
2. Each ray is set up according to the given camera parameters and respective pixel position.
3. Each ray is traversed by stepping along it to accumulate pixel color and opacity.
4. Image pixels are displayed on the screen.

Our GPU-based implementation effectively supports two main modes of working:

- *Exploration mode*: This mode provides a set of fundamental real-time interaction operations, such as rotation, translation and zoom. These operations cause recomputation of steps 2–4 in the GPU, which can be done in real time.
- *Editing mode*: In this mode, the mouse can be used primarily as an eraser, sculpting tool, or drill, while the

Fig. 6 Intuitive sketch-based visualization techniques with precise feedback information



middle and right buttons can still be used for translation and zoom. Volume data selected by the eraser are erased by modifying the transfer function, changing the opacity to 0.0. We directly use the eraser, proposed by [38]. The sculptor works in image space, and can cut off unwanted voxels from components by using a local-detail carving operation. We provide two formed (square and circle) shapes as Fig. 7 left, where the opacity $o_{i,j,k}$ of point (i, j, k) is computed relative to the sculptor center (i_c, j_c, k_c) as:

$$o_{i,j,k} = ((1 - h^2) \times ((i - i_c)^2 + (j - j_c)^2)) / \delta_s^2 \quad (6)$$

where $h \in [0, 1]$ is the hardness of sculptor, and $\delta_s \in [1, 10]$ is the size of sculptor. The driller works in voxel space, and allows the user to see the kernel of a component directly. The driller also has two forms: cube and

sphere as Fig. 7 right. The opacity $o_{i,j,k}$ of point (i, j, k) is computed relative to the driller center (i_c, j_c, k_c) as:

$$o_{i,j,k} = ((1 - h^2) \times ((i - i_c)^2 + (j - j_c)^2 + (k - k_c)^2)) / \delta_d^2 \quad (7)$$

where $h \in [0, 1]$ and $\delta_d \in [1, 100]$ are the hardness and size of driller, respectively.

As shown in Fig. 6, information for precise interaction is updated by frame, including space position, window position, color, and data value of interaction pointer. On the other hand, to get graceful label rendering result, we use rules from [5] to ensure that annotations do not overlap, connecting lines between annotations and anchor points do not cross, annotations do not occlude any other structures, and annotations are placed as close as possible to their anchor points.

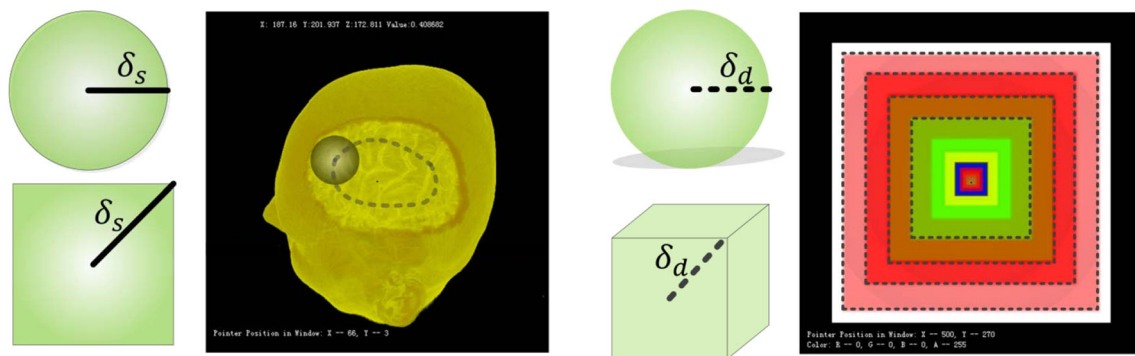
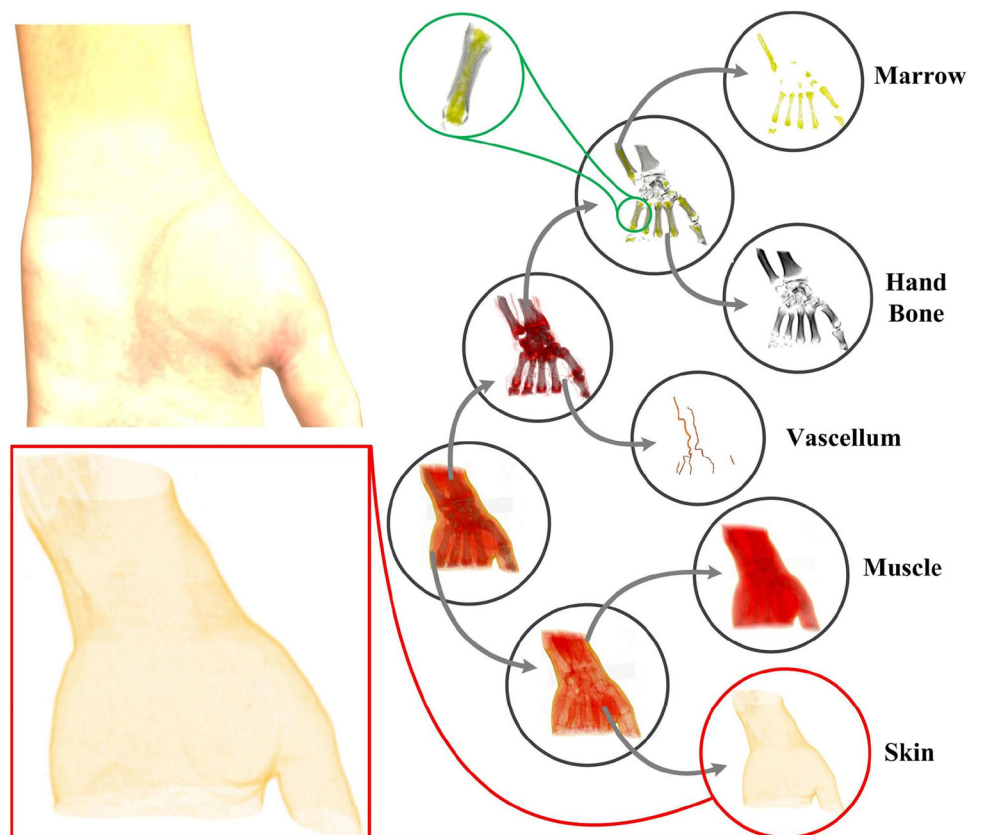


Fig. 7 Sculptor (*left*) and driller (*right*) tools

Fig. 8 Semi-automatic hierarchical segmentation. *Top-left* the provided model, *bottom-left* magnification of the rendered hand's skin



5 Results

5.1 Experimental results

We implemented our method and its application by using CUDA 5.5 and tested it with several volume data on a PC with Intel 2.80 GHz Core i7, 12 GB DDR3 RAM and GeForce GTX 580.

In our paper, we tested our method with representative six datasets with the help of human anatomy model (HAM). These datasets not only range from part or parts (like foot, hand, head, and abdomen) to whole (like skeleton and body surface), but also range from simple (like foot, hand) to com-

plex (like head, abdomen). Since most of the medical volume datasets that come from (parts of) human body and the HAM is widely regarded as fundamental model, we make use of HAM model by now.

Our method could easily get hierarchical structure of a multicomponent volume as shown in Fig. 8, in which we highlight hierarchical segmentation of a wrist dataset. The result shows that our model-inspired segmentation is simple but effective. Segmentation by analogical inference allows us to apply clear rendering to anatomical meaningful components directly. The wrist skin at the bottom left achieved with our method is comparable to the model at the top left.

Fig. 9 Semi-automatic annotation example via partial matching

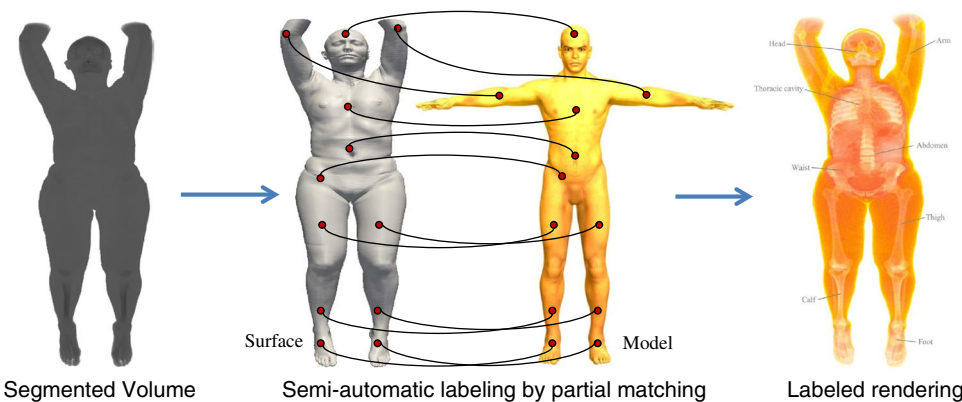


Fig. 10 Semi-automatic annotation examples. *Left* semantic model, *right* labeling result

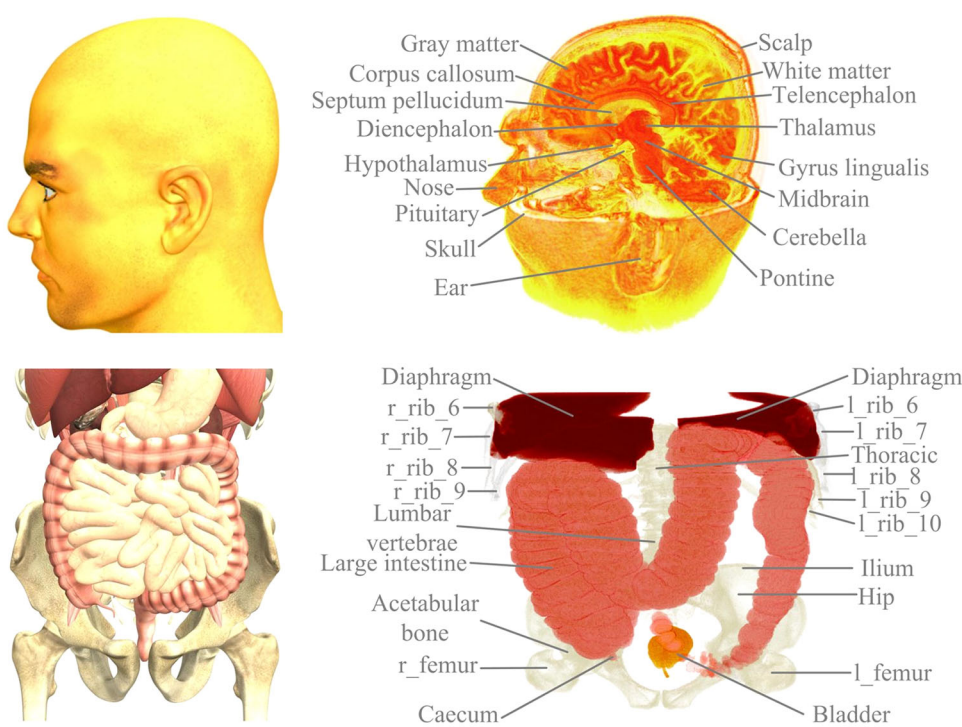


Figure 9 demonstrates the inference of whole body structure from the given anatomical model by use of partial matching. Only salient components are labeled in the example, and the partial matching process could be allowed to complete fully automatically without any user interaction. However, for complex cases (e.g. Figs. 4, 10), user interaction may be needed during correspondence building.

Table 1 lists related information of the volume datasets tested, namely dimension, number of components, number of nodes in the corresponding constructed graph, number of nodes in the corresponding abstracted graph. Table 2 records the labeling accuracy (without and with one correspondence pair indicated by the user) and the time taken for various examples including those illustrated(Figs. 4, 9, 10). The time cost records the execution time (in seconds) to perform the partial matching. Given one correspondence pair, this is the matching time after correspondence assignment. The exper-

Table 1 Related information of volumes

Examples	Dimension	N_c	N_n	N_{an}
Figure 3 foot	$256 \times 512 \times 128$	9	9	9
Figure 4 skeleton	$512 \times 512 \times 1,871$	22	26	10
Figure 8 hand	$256 \times 256 \times 128$	15	21	12
Figure 9 body surface	$512 \times 512 \times 1,871$	12	12	7
Figure 10 head	$512 \times 512 \times 356$	17	25	17
Figure 10 abdomen	$512 \times 512 \times 512$	24	26	20

N_c number of components in the volume, N_n number of nodes in the corresponding constructed graph, N_{an} number of nodes in the corresponding abstracted graph

imental results make clear that a few interactions can help improve the accuracy ratio and speed up the matching procedure.

Table 2 Labeling statistical records on accuracy and time cost (in seconds)

Examples	w/o interaction		w/1 correspondence pair	
	Accuracy (%)	Time	Accuracy (%)	Time
Figure 3 foot	84	0.3	96	0.2
Figure 4 skeleton	88	2.2	95	1.9
Figure 8 hand	100	0.2	–	–
Figure 9 body surface	100	1.4	–	–
Figure 10 head	76	4.6	89	3.2
Figure 10 abdomen	83	4.2	92	3.1

5.2 Discussion

We have evaluated our method by gathering feedback in two separate user evaluations with different target audiences. The first evaluation targeted medical domain experts, while the second evaluation examined the usefulness of our method for recent medical graduates. Both elicited qualitative assessments of the benefits and drawbacks of our method.

The participants in the first evaluation were a neuro-ophthalmologist in the first session and a group of eight radiologists in the second session. Each session comprised an initial demonstration and explanation of the system, guided experimentation by the participants, and a concluding semi-structured interview. The feedback of the medical experts was positive in general. Besides, they also affirmed the efficiency and accuracy of our method as shown in Table 2 they quickly understood the interface and were able to use it after a short period of instruction.

The participants in the second evaluation were ten graduate medical students, who could understand and use our system quickly. These students particularly liked the fact that when using our approach, they could transfer knowledge and skills from the semantic model, and that it gave them useful cues to rendering and annotation.

One difficulty of our method is the balance between automation and robustness of using the reference semantic relationships as a basis for analogical inference, and subsequent effects on interactive editing. Expert users often wish to edit the visualizations produced, e.g. by modifying transfer functions. Novices, on the other hand, tend to approach creation of an image in a less interactive way, may be because illustration based on real data has not been possible to a large degree until recently. They found it difficult to perform segmentation, specify annotation, and assign proper color mappings for anatomical organs.

6 Conclusions

In summary, we propose a novel multicomponent volume exploration method that makes use of admitted geometry model to help volume segmentation, labeling, and rendering.

In our approach, GMM first classifies the volume, region-grown then segments each class into components. Labeling is deduced from correspondences between the model and the segmented volume. Last, realistic rendering is induced with color value from the model, and some essential precise sketch-based interactive visualization techniques are constructed for further exploration. Our method takes exiting admitted knowledge into volume exploration which greatly reduces the amount of user interaction required for many time-consuming and laborious tasks which are also prone to errors and omissions, such as volume segmentation, labeling and realistic rendering. To our knowledge, existing related works mainly cope with volume labeling problem by interaction, while our method achieves automatic labeling by learning labeling and rendering information from existing model.

Our method could be used in several applications: in medical education, novice could study body easier and quicker; in disease diagnosis, doctor could quickly get specific information (like name, position, visual form) of abnormal organ that could remarkably raise diagnosis efficiency and accuracy. In the future, we are planning to test and verify these applications in practice. Moreover, it is also indispensable for inspiring some new but promising applications, for example, volume searching which finds certain part of volume data by keywords, which is parallel to searching image in Google picture or 3D model in Google 3D Warehouse.

In the following, we will emphasize on raising the accuracy of volume-model correspondence by taking more information, like shape and structuring information between the components as in [15], into account. We will also provide a user-interface, which manually controls the opacity of the segmented regions. In addition, exploring more potential applications like model-driven analysis of volume data of animals and model-driven feature analysis in Computational Fluid Dynamics is our future work.

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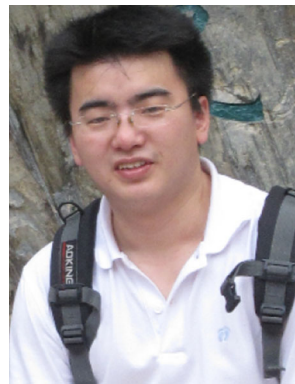
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